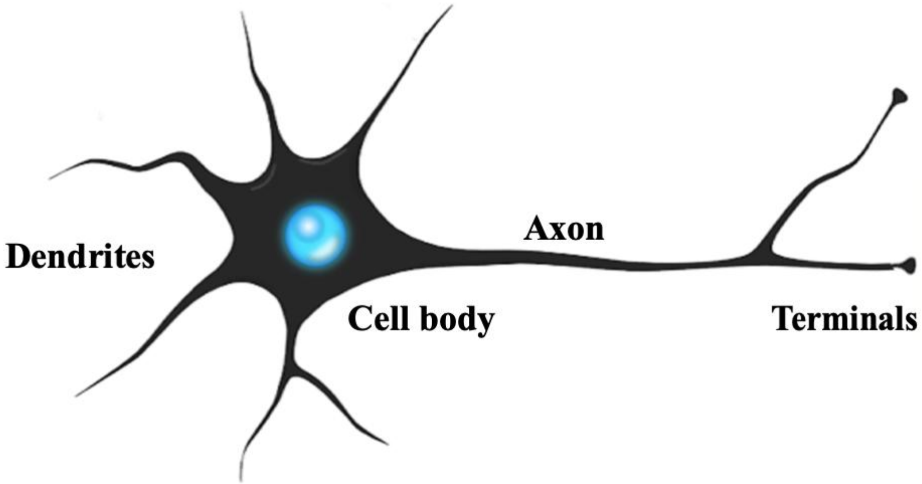
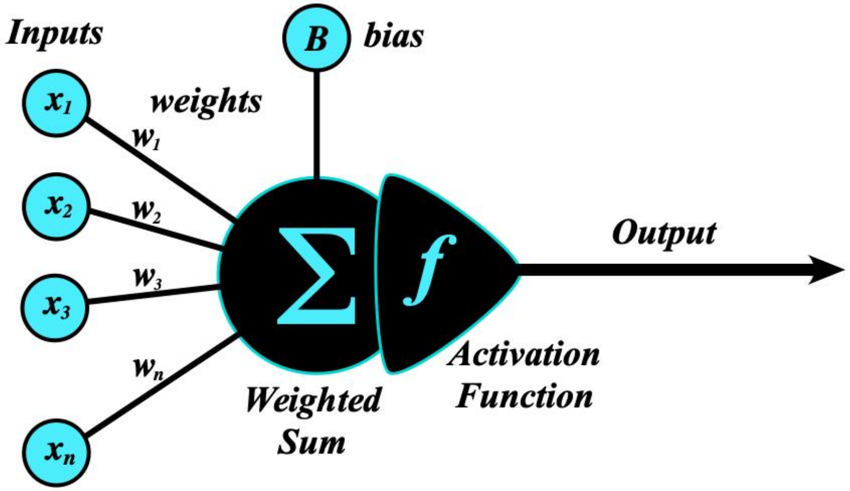
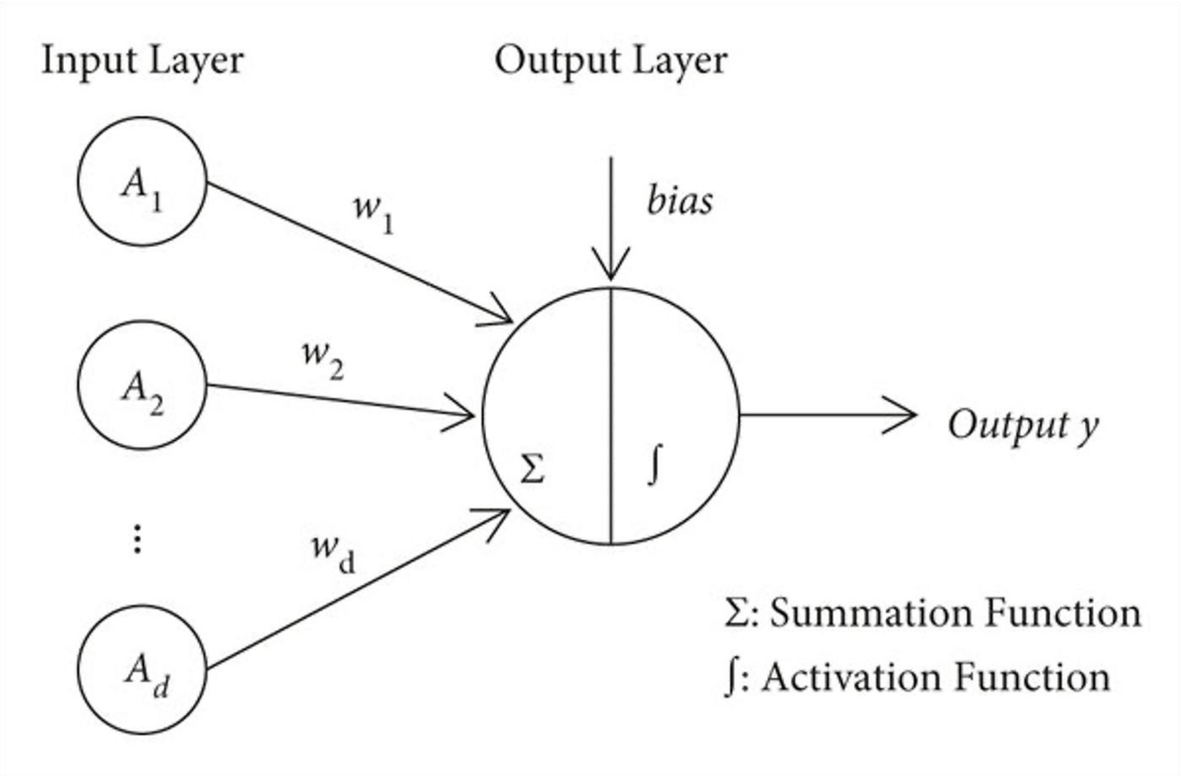
Module 1: The Artificial Neuron (Perceptron)

- Welcome to Phase 5: Deep Learning, Computer Vision & LLMs!
- Part 1 focuses on The Foundations of Artificial Neural Networks (ANN).



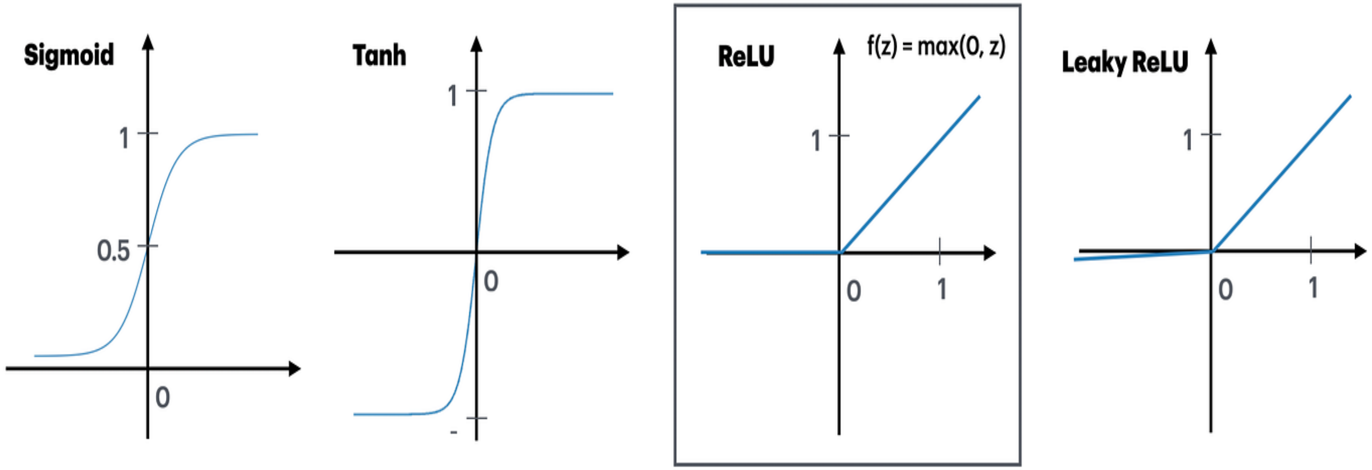


The Perceptron: The Core Building Block



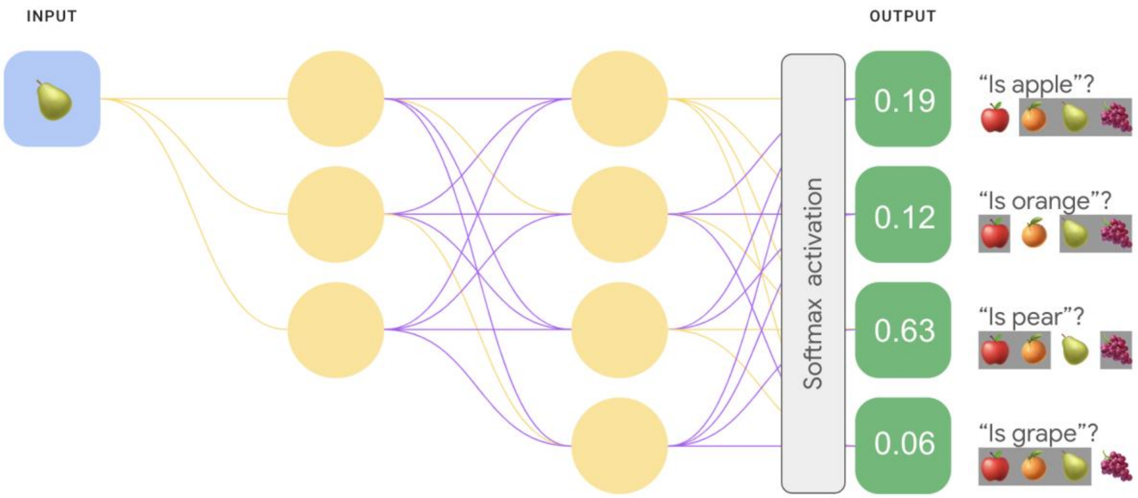
- The Perceptron is the foundational unit of deep learning.
- It requires understanding **Weights**, which represent the "importance" of incoming data.
- It utilizes **Biases**, which act as "thresholds" for the neuron to activate.
- The neuron processes these inputs using **summation math**.

Activation Functions: Introducing Non-Linearity



- Activation functions are critical for introducing non-linearity into the network, allowing it to learn complex patterns.
- Traditional networks often use the $Sigmoid$ and $Tanh$ functions.
- Modern deep learning primarily relies on $ReLU$ and $Leaky\ ReLU$ functions.

The Softmax Function: Multi-Class Decisions



- The Softmax function is typically applied at the very end of a neural network.
- Its primary job is converting raw scores (logits) into probabilities.
- This makes it essential for multi-class decisions (e.g., deciding if an image is a cat, dog, or bird).